COMPUTER SCIENCE MSC CURRICULUM

Program: Artificial Intelligence Specialization

ASSESSMENT NOTE:
*** Subjects are offered in Autumn (A), Spring (S), or both semesters (A,S)**
**** Core subjects regardless of specialization**
***** Students must complete 47 credits from compulsory electives**

=== COMPULSORY SUBJECTS (37 Credits) ===
IPM-22fRMEG: Research methodology
Lecture + Practice | 1 Credit | Semesters: 2 | Offered: A,S | 5 Credits total
IPM-22fASTE: Advanced Software Technology (Lecture) **
2 Lecture, 0 Practice | 4 Credits | Semesters: 2 | Offered: S
IPM-22fDAAE: Design and analysis of algorithms (Lecture) **
2 Lecture, 0 Practice | 4 Credits | Semesters: 2 | Offered: S
IPM-22fmiDNDEG: Deep Network Developments
2 Lecture, 2 Practice | 6 Credits | Semesters: 1,2 | Offered: A,S
IPM-22fmiMTAAEG: Methods and tools for AI applications
2 Lecture, 2 Practice | 6 Credits | Semesters: 1 | Offered: A
IPM-22fmiPAIEG: Principles of artificial intelligence
2 Lecture, 2 Practice | 6 Credits | Semesters: 1 | Offered: A
IPM-22fmiDRLEG: Deep Reinforcement Learning
2 Lecture, 2 Practice | 6 Credits | Semesters: 2 | Offered: S
Prerequisite: IPM-22fmiDNDEG

COMPULSORY ELECTIVE SUBJECTS (Select 47 Credits)
--- FOUNDATIONAL COURSES ---
IPM-22fmiIDSEG: Introduction to Data Science
2 Lecture, 2 Practice | 6 Credits | Semesters: 1,2 | Offered: A,S
IPM-22fmiTAMEG: Topics in Applied Mathematics
2 Lecture, 2 Practice | 5 Credits | Semesters: 1,2 | Offered: A,S
IPM-22fmiPREPG: Preparation course for master studies and developing learning skills
0 Lecture, 3 Practice | 2 Credits | Semesters: 1,2 | Offered: A,S

*** Mandatory for international students**
--- COMPUTER VISION & PERCEPTION ---
IPM-22fmiCVEG: 3D Computer Vision
2 Lecture, 2 Practice | 6 Credits | Semesters: 1 | Offered: A
IPM-23fmi3DCPAEG: 3D point cloud processing and analysis
2 Lecture, 2 Practice | 6 Credits | Semesters: 1 | Offered: A

--- MACHINE LEARNING & DEEP LEARNING ---
IPM-22fmiMLEG: Machine Learning
2 Lecture, 2 Practice | 6 Credits | Semesters: 2 | Offered: S
Prerequisite: IPM-22fmiIDSEG

IPM-22fmiAMLEG: Advanced Deep Network Development
2 Lecture, 2 Practice | 6 Credits | Semesters: 1 | Offered: A
Prerequisite: IPM-22fmiDNDEG

--- NATURAL LANGUAGE PROCESSING ---
IPM-23fmiNLPFMEG: Natural Language Processing and Language-based Foundation Models
4 Lecture, 4 Practice | 12 Credits | Semesters: 1,2 | Offered: A,S
Prerequisite: IPM-22fmiDNDEG

--- AI SYSTEMS & APPLICATIONS ---
IPM-22fmiCOSCEG: Cognitive Science
2 Lecture, 2 Practice | 6 Credits | Semesters: 2 | Offered: S
IPM-22fmiGTEG: Game theory
2 Lecture, 2 Practice | 6 Credits | Semesters: 2 | Offered: S
IPM-22fmiLPE: Logic programming (Lecture)

2 Lecture, 0 Practice | 3 Credits | Semesters: 2 | Offered: S
IPM-22fmiLPG: Logic programming (Practice)
0 Lecture, 2 Practice | 3 Credits | Semesters: 2 | Offered: S
* Practice part is prerequisite for lecture grade
IPM-22fmiMASEG: Multi-agent systems
2 Lecture, 2 Practice | 6 Credits | Semesters: 2 | Offered: S
IPM-22fmiACEG: Affective computing
2 Lecture, 2 Practice | 6 Credits | Semesters: 1 | Offered: A
Prerequisite: IPM-22fmiDNDEG
IPM-22fmiROBEG: AI Robotics
2 Lecture, 2 Practice | 6 Credits | Semesters: 1 | Offered: A
IPM-22fmiCOLLIEG: Collective Intelligence
2 Lecture, 2 Practice | 6 Credits | Semesters: 1 | Offered: A
IPM-22fmiCIEG: Computational Intelligence
2 Lecture, 2 Practice | 6 Credits | Semesters: 1 | Offered: A
IPM-22fmiEIEG: Embodied Intelligence
2 Lecture, 2 Practice | 6 Credits | Semesters: 1 | Offered: A
IPM-23fmiSJEG: Statistics for signal processing
2 Lecture, 2 Practice | 6 Credits | Semesters: 1 | Offered: A

--- PROJECT COURSES ---
IPM-22fmiAIPLAB1: AI Project Lab I
0 Lecture, 2 Practice | 4 Credits | Semesters: 1 | Offered: A
IPM-22fmiAIPLAB2: AI Project Lab II
0 Lecture, 4 Practice | 6 Credits | Semesters: 1 | Offered: A
* Fulfills internship requirement (6 weeks, 240 hours)

=== THESIS & ELECTIVES ===
IPM-22fTHCONS: Thesis consultation
30 Credits | Semesters: 3,4 | Offered: A,S
Elective Subjects: 6 Credits

=== EIT DIGITAL MASTER PROGRAMME (Optional) ===
IPM-22fI&EBEG: I&E Basics
IPM-22fI&EBDL1G: Business Development Lab I
IPM-22fI&EBDL2G: Business Development Lab II
IPM-22fI&EIAOEEG: Innosocial aspects of entrepreneurship
IPM-22fI&ETSSG: Thematic Summer Schools with I&E project
IPM-22fI&ESTEG: I&E Study
* Mandatory for EIT students, elective for others

=== CREDIT SUMMARY ===
Compulsory Subjects: 37 Credits
Compulsory Electives: 47 Credits
Elective Subjects: 6 Credits
Thesis: 30 Credits
TOTAL DEGREE CREDITS: 120 Credits

Semester Credit Distribution:
Semester 1: 30 | Semester 2: 30 | Semester 3: 30 | Semester 4: 30